

DIVISION S-7—FOREST & RANGE SOILS

Morphological Characteristics of Macropores and the Distribution of Preferential Flow Pathways in a Forested Slope Segment

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ABSTRACT

Water flow through soil macropores is important in determining hydrologic responses in forested watersheds. Morphological characteristics of macropores and distribution of preferential flow pathways were evaluated in a forest hillslope segment using a combination of staining agents. Almost 80% of described macropores were roughly elliptical with eccentricities ranging from 0.256 to 0.998 (mean of 0.652) and lengths ranging from 2.0 to 61.8 cm (mean of 11.6 cm). Tortuosity of macropores tended to increase with increasing length up to about 30 cm, with a mean value of 1.14. Macropores were aggregated in large clumps within the soil profile. Living and decayed roots and associated vertical zones of loose soil and humus contributed to preferential flow pathways in this soil. Subsurface flow patterns, detected by upslope injection of dilute white paint solution, showed a strong interaction between the soil matrix and macropores. Subsurface flow was lateral along the bedrock and between A and B horizons, with a perched water table occurring on sections of both. Dye tests also showed that flow occurred within surface bedrock fractures. This fracture flow was sometimes connected to macropores through zones of local wetness. Thus, we conclude bedrock topography and fracture characteristics may contribute significantly to preferential flow pathways at the hillslope scale. Even though individual macropores were rather short, the coupling of these flow paths with the soil matrix, bedrock fractures, living and decayed roots, and perched water tables produced complex networks of interconnected preferential flow pathways, all of which help explain the stormflow response observed in the catchment.

THE IMPORTANCE of soil macropore and pipe flow is generally acknowledged especially in forest soils (Beven and Germann, 1982). Large storm discharge from soil macropores and pipes have been measured at several sites (e.g., Jones, 1978; Tanaka et al., 1988; Tsukamoto and Ohta, 1988; Kitahara and Nakai, 1992). Studies of macropore-pipe flow mechanisms have been conducted in field settings (e.g., Luxmoore et al., 1990; Tsuboyama et al., 1994b; Sidle et al., 1995b) and in laboratory experiments (e.g., Germann and Beven, 1981; Ela et al., 1992; Sidle et al., 1995a; Terajima et al., 1996a). Flow pathways in soils can also be estimated or inferred by hydrologic models (e.g., Jones, 1988; Germann, 1990; Tani, 1997).

The Maimai research catchment in New Zealand has been the site of ongoing hillslope research by several

research teams since the late 1970s (e.g., Mosley, 1979; Sklash et al., 1986; McDonnell, 1990). Different views emerged from these studies concerning the importance and mechanisms of soil macropore and pipe flow. Mosley (1979) concluded that subsurface flow through macropores was the primary contributor to storm runoff. Sklash et al. (1986) evaluated the roles of old and new water using natural isotopes and concluded that stormflow was largely composed of old stored water. From this they inferred that macropore flow was not significant. McDonnell (1990) noted that invading new water perched at the soil-bedrock interface and backed up into the matrix where it mixed with a much larger volume of stored soil water. None of these studies measured morphological characteristics of soil macropores needed to develop and modify perceptual models of subsurface flow in hillslopes.

Dyes can be used to investigate preferential flow pathways including macropores and pipes through soil profiles. Dye movement has been used to evaluate flow down worm channels (Ehlers, 1975; Omoti and Wild, 1979; Van Stiphout et al., 1987), along vertical ped faces and pipes (Bouma and Dekker, 1978; van Stiphout et al., 1987; Wang et al., 1993) and along living and decayed root channels (Mosley, 1979; Noguchi et al., 1997a).

Recent studies have evaluated subsurface flow through the soil matrix and macropores of a hillslope segment, piezometric response in a zero-order basin, and hydrologic response at various spatial scales in a catchment at the Hitachi Ohta Experimental Watershed (Tsuboyama et al., 1994a, b; Sidle et al., 1994, 1995b). In this same catchment, Noguchi et al. (1997b) evaluated morphological characteristics of macropores (density, diameter, direction, gradient, and origin) in forest soil profiles. The objectives of this study were to evaluate additional morphological characteristics of various macropores (shape, length, connectivity, tortuosity, and distribution) that relate to their hydrologic function, as well as to describe the three-dimensional distribution of preferential flow pathways in a forested slope segment using two staining techniques.

MATERIALS AND METHODS

The study was conducted at the Hitachi Ohta Experimental Watershed, on the east side of Honshu (the Mainland) of Japan at 36° 34' N and 140° 35' E. Based on an 11-yr record at the site, mean annual precipitation is 1485 mm. Prior to the 20th Century, the watershed was covered with a natural hardwood forest. In the early 1900s, clearcutting began in various forest blocks. Cutover sites were replanted with sugi [*Cryptomeria japonica* (L.F.) D. Don] and hinoki [*Chamaecyp-*

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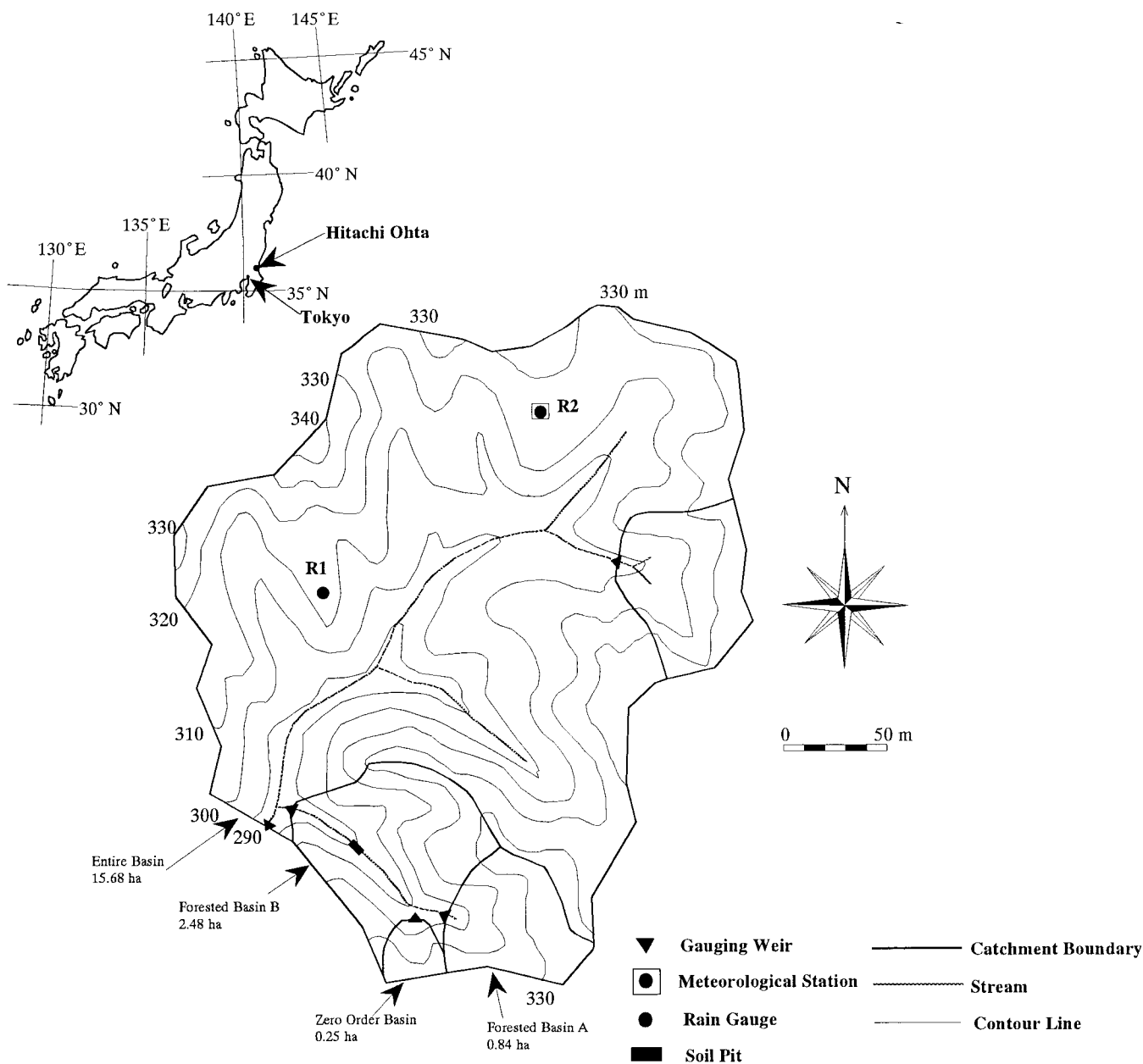


Fig. 1. Map of the Hitachi Ohta Experimental Watershed showing the locations of basins, their instrumentation, and the investigation site.

paris obtusa (Siebold & Zucc.) Endl.] around 1920. Hardwood trees and various understory species coexist in gaps within the conifer forest. Soils are derived from volcanic ash and are classified as Inceptisols. Surface and subsurface soil textures based on the International Society of Soil Science system used in Japan (Yong and Warkentin, 1966) are clay loams. Surficial geology is metamorphic rock, primarily schist and amphibolite. An experimental soil pit was excavated at the base of the hillslope about 60 m upstream of the weir at forested basin B (Fig. 1). The equivalent USDA classification at the soil pit is Lithic Dystrochrepts. The hillslope was 49 m long with an average gradient of 39°. Characteristics of the research site have been described in detail by Tsuboyama et al. (1994b) and Sidle et al. (1995b).

Soil Physical Properties

Five intact core samples (100 cm² area, 4 cm long) were collected from the A and B horizons beside the pit. The intact

samples were analyzed for saturated hydraulic conductivity (K_{sat}) by the constant-head method (Mashimo, 1960), water retention at low pressure by the soil column method and at high pressure by pressure chamber method (Hasegawa et al., 1997). Bulk density was determined by the core method (Blake and Hartge, 1986). Bulk samples were also collected from the same horizons for particle-size analysis by the pipette method (Gee and Bauder, 1986).

Experimental Procedures

Because some dyes (e.g., methylene blue) exhibit high adsorption in organic-rich soils, they are not particularly useful to delineate flow pathways. Instead, we used a dilute white paint solution (acrylic fiber resin emulsion) that has been found to have minimal adsorption in forest soils (Tsuji-mura et al., 1991; Noguchi et al., 1997a) in our study to trace flow pathways into the soils. Additionally, we conducted compari-

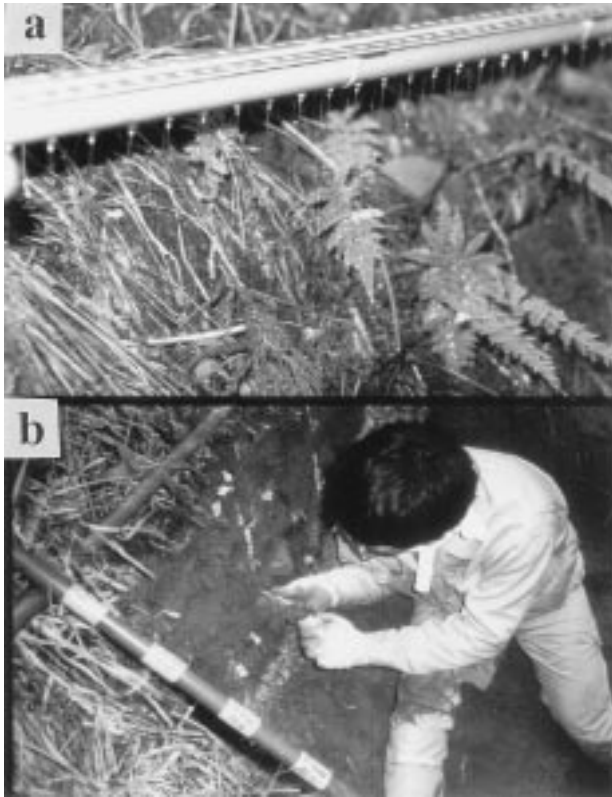


Fig. 2. Two staining techniques: (a) white paint as dye sprinkled using irrigation system; (b) powder chalk sprayed into the macropores.

sons between white paint dye and methylene blue dye in small-scale infiltration tests at a nearby site in the watershed with the same soils. Test results showed that methylene blue was highly adsorbed by the organic horizons of the soil, while white paint readily infiltrated through the organic horizon and into the underlying mineral soil. Maximum particle size in the white paint was 10 μm .

Prior to dye application, water was applied at a rate of 60 L h^{-1} for 2 h at a 2-m line irrigation source of intravenous needles (0.7 mm i.d.) positioned 2 m upslope (slope distance)

from the pit face (Fig. 2a). Some 96 needles were arranged along two closely spaced rows with the needle orifices positioned 2 to 39 cm (weighted mean of 22 cm) above the soil surface. At this point, steady flow from the pit face was observed. The dilute paint solution (20% paint by volume) was then injected at a constant rate for 1 h, allowing the paint solution to emerge from the face of the pit. Details of the irrigation system are presented by Tsuboyama et al. (1994b).

Diameters used to define macropores have differed widely in the literature (Luxmoore et al., 1990). We use the term to including all pores, including pipes with diameters ≥ 2 mm and length ≥ 20 mm. Both small and large diameters (two axes) for all macropores were measured. We visually observed all of these macropores in the excavated soil profiles. The depth distribution of each soil layer (organic-rich layer consisting of O and A horizons, and B horizon) and the position of macropores were measured using a 10-cm grid. Distribution patterns of dye in the soil profile were photographed. In addition, the continuity of macropores was traced to the next excavated soil profile by spraying powdered chalk (red, blue, and yellow) into the macropores at the pit face (Fig. 2b). Next, an 8-cm vertical slice of soil was excavated from the pit face and the recovery of the powdered chalk through this 8-cm slice was noted. Each new pit face was carefully prepared with a knife to preserve the ped and macropore structure. The process was repeated 20 times in the upslope direction, so that the entire distance from the original pit face to the line irrigation source was mapped. Also, for each successive excavation the distribution of white paint, including interactions and associations with macropores, was measured.

Eccentricity of Macropore

Eccentricity of macropores (Ecc) was calculated as follows:

$$Ecc = [(d_l/2)^2 - (d_s/2)^2]^{0.5} / (d_l/2) \quad [1]$$

where d_l and d_s are large and small diameters of macropores, respectively.

Tortuosity of Macropore

Tortuosity of macropores (T) was calculated as follows:

$$T = \sum l_i / L \geq 1 \quad [2]$$

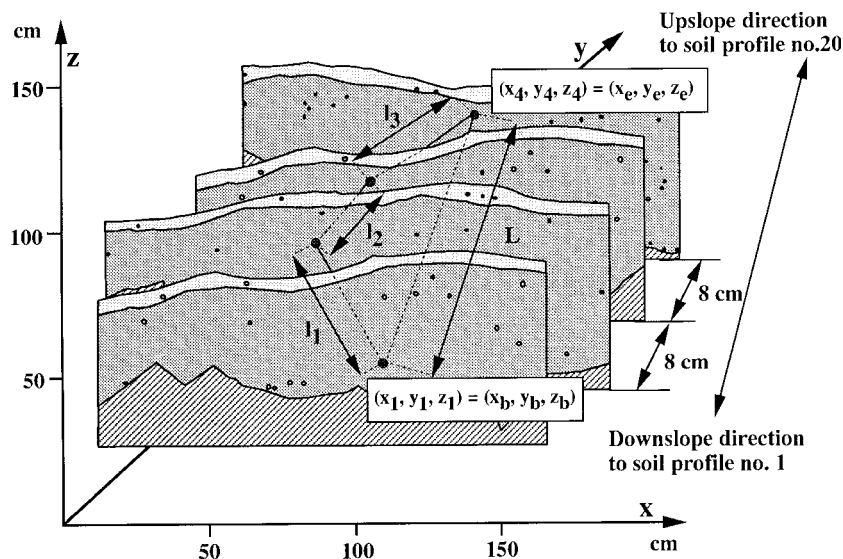


Fig. 3. Schematic of tortuosity for macropores in soil profile.

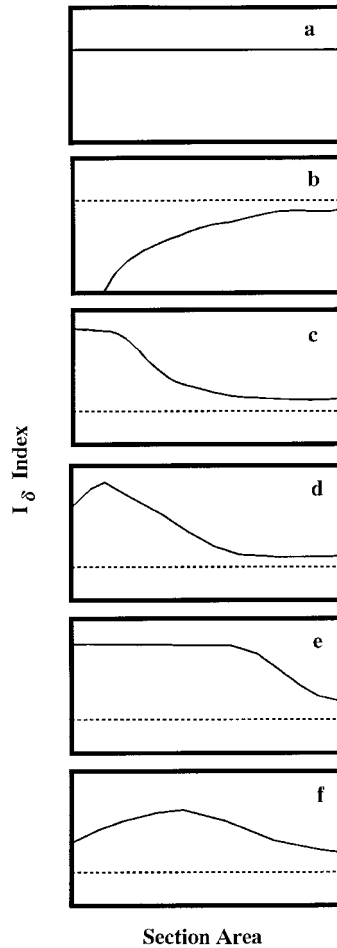


Fig. 4. Schematic representations of I-area size relations for various distributional patterns. The broken lines indicate the value of unity. (a) Random distribution; (b) uniform distribution; (c) aggregated distribution with small clump or clumps (intra-clump distribution is at random); (d) aggregated distribution with small clump or clumps (intra-clump distribution is uniform); (e) aggregated distribution with large clump or clumps (intra-clump distribution is at random); (f) aggregated distribution with large clump or clumps (intra-clump distribution is uniform).

and

$$\sum_{i=1}^{n-1} l_i = \sum_{i=1}^{n-1} [(x_i - x_{i+1})^2 + (y_i - y_{i+1})^2 + (z_i - z_{i+1})^2]^{0.5} \quad [3]$$

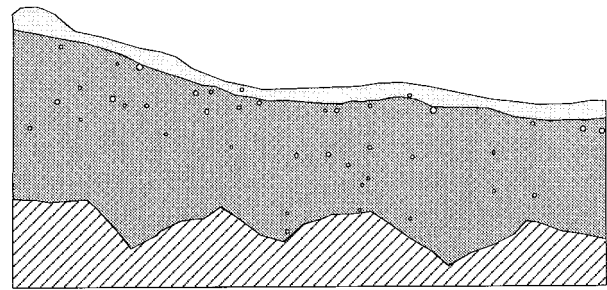
$$L = [(x_e - x_b)^2 + (y_e - y_b)^2 + (z_e - z_b)^2]^{0.5} \quad [4]$$

where L is the linear distance from the beginning to the end of each macropore, l is distance along the actual path of the macropore from one soil profile to next soil profile, and n is the number of soil profiles (≥ 3) through which macropores persisted (Fig. 3). Because many macropores were <16 cm in length, only 29 could be included in tortuosity calculations. These values under-represent actual tortuosities, since they are the sum of straight line segments through three or more soil cross sections.

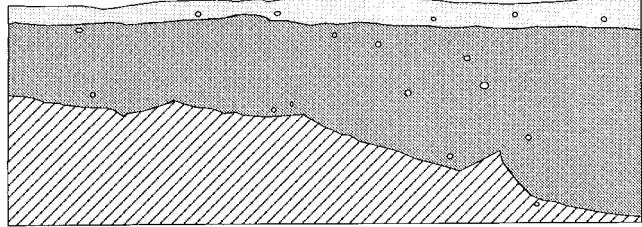
Evenness of Distribution of Macropores

Morisita (1959) developed an index to analyze the evenness of the spatial distribution of individuals in a population. This methodology has been used to evaluate the spatial patterns of plant and insect species (Morisita, 1962; Akashi, 1996) as

Profile no. 17



Profile no. 11



Profile no. 1

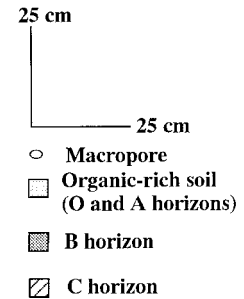
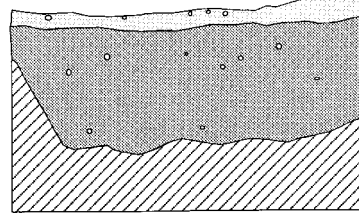


Fig. 5. Some examples of schematics of distribution of macropores.

well as soil pipes (Kitahara et al., 1988). The spatial distribution of macropores in our soil profiles was analyzed by Morisita's I_δ Index:

$$I_\delta = q \frac{\sum_{i=1}^q n_i(n_i - 1)}{N(N - 1)} \quad [5]$$

$$N = \sum_{i=1}^q n_i \quad [6]$$

where n_i is the number of individuals in each section ($i = 1, 2, 3, q$) and q is the number of sections of a certain grid size within each profile. Morisita (1959) developed characteristic relationships between I_δ and grid size which are useful for determining spatial distributions of individual (Fig. 4).

The significance of the departure from randomness of the distribution was tested by comparing the F_0 value with the value of F_{α}^{q-1} where F_0 is defined as

$$F_0 = \frac{I_\delta(N - 1) + q - N}{q - 1} \quad [7]$$

Description of Soil Pit

Twenty successive vertical profiles ranging in width from 100 to 185 cm were excavated in the upslope direction. Examples of the distribution of soil horizons and macropores in three profiles are shown in Fig. 5. Soil depths varied considerably in most pits; mean depths of organic-rich soil (O and A horizons) and B horizon in each profile ranged from 2.6 to 6.8 cm (overall mean 4.5 cm) and from 12.2 to 42.2 (overall

Table 1. Morphological characteristics of soil macropores (from Noguchi et al., 1997b.)

Horizon	Diameter					Gradient†				Direction‡			
	Min.	Max.	Mean§	Mean	S.D.	Min.	Max.	Mean§	S.D.	Min.	Max.	Mean§	S.D.
	mm					degree				degree			
A	2	48	20.2	18.1	8.6	−66	36	−26.0	17.3	−80	84	−7.3	33.7
B	3	60	14.2	11.2	9.7	−72	67	−14.9	23.4	−90	80	−2.3	35.7

† The gradient of soil macropores was defined as 0 for orientation perpendicular to the pit face, with the angles in the upper and lower directions assigned negative and positive values, respectively.

‡ The direction of macropores was defined as 0 for the azimuth angle of the pit face. Clockwise and counterclockwise orientations were assigned positive and negative values, respectively.

§ Arithmetic mean.

|| Geometric mean.

Table 2. Physical properties of the A and B soil horizons used in the study of water flow through macropores.

Horizon	Sand (0.02–2 mm)	Silt (2–20 μm)	Clay (<2 μm)	Bulk density†	$K_{sat}‡$	Water content at pressure heads (MPa)					
						0	−0.0031	−0.0098	−0.0309	−0.0491	−0.0980
	%			$g\ cm^{-3}$	$m\ s^{-1}$	$m^{-3}\ m^{-3}$					
A	50.3	33.6	16.1	0.690	2.9×10^{-4}	0.765	0.503	0.445	0.388	0.360	0.323
B	47.3	36.8	19.5	0.857	3.1×10^{-4}	0.695	0.421	0.374	0.324	0.304	0.271

† Arithmetic mean of three samples.

‡ Geometric mean of five samples.

Table 3. The range in sizes of macropores in this study was substantially greater than that reported in a 1989 study by Kitahara. The *n* denotes sample number.

Study	Dimensions	Minimum	Maximum	Mean	SD
Profiles no. 1–20 <i>n</i> = 453	Area (cm ²)	0.03	28.27	2.64	3.06
	Large diameter (cm)	0.2	7.0	1.83	1.16
	Small diameter (cm)	0.2	6.0	1.36	0.87
	Eccentricity (e)†	0.256	0.998	0.652	0.166
	Hydraulic radius (cm)	0.05	1.50	0.377	0.236
Kitahara (1989) <i>n</i> = 9	Area (cm ²)	0.82	5.89	1.06	1.97
	Avg. max. diameter (cm)	1.17	3.94	2.63	0.90
	Avg. min. diameter (cm)	0.74	2.76	1.94	0.62
	Eccentricity	0.393	0.836	0.626	0.158
	Hydraulic radius (cm)	0.268	0.667	0.522	0.143

† Eccentricity could not be calculated when the large and small diameters were equal. Excluding those, eccentricities were calculated on *n* = 359.

mean 27.3 cm), respectively. Macropore densities ranged from 5.2 to 29.1 (mean 14.1) per m of profile width and from 16.8 to 75.1 (mean 43.0) per m² of profile area. The O horizon was very thin (<1 cm).

The diameters, gradients, and directions of described macropores are shown in Table 1 (Noguchi et al., 1997b). The mean macropore diameter in the O + A horizons was larger than in the B horizon. The dominant gradients of macropores were at negative oblique angles. Approximately 90% of the macropores in the O + A horizon and approximately 80% of the macropores in the B horizon fell within the range between −50 and 50 degree planar direction. Most macropores were formed by subsurface erosion (evidence of water flow or actual water flow observed in the macropores), root channels, and interactions between subsurface erosion and root channels (Noguchi et al., 1997b).

RESULTS AND DISCUSSION

Physical Properties

General properties of the A and B horizons are shown in Table 2. There were no significant differences for particle size distribution and K_{sat} between A and B horizons. The bulk density of A horizon was lower than that of B horizon. Water content of A horizon was larger than in the B horizon throughout the range of all tested pressure heads. However, the coarse pore space (between 0 and −0.003 MPa) and the fine pore space (be-

tween −0.003 and −0.098 MPa) percentages were almost the same for A and B horizons.

Morphological Characteristics of Macropores

About 80% of described macropores were elliptical with eccentricities ranging from 0.256 to 0.998. This range in values was wider than the range measured in soil pipes by Kitahara (1989) (Table 3). Macropores formed by root channels comprised 70% of all macropores in the organic-rich layer and 55% in the B horizon (Noguchi et al., 1997b). This origin of macropores may contribute to their elliptical shape. It was observed that the long axis of the decayed root macropores was typically oriented parallel to the soil surface. In contrast, live roots were typically circular. We believe that this elliptical shape and orientation of decayed root macropores may result from compression and collapse as roots decay. Similar results have been observed in forest soils in Hokkaido, Japan (Kitahara et al., 1988; Kitahara, 1989).

About 70% of the macropores were discontinuous (dead end) after one 8-cm section and only 1% of all macropores continued through five soil sections (i.e., 40 cm). Actual length of macropores estimated as the sum of the straight line segments through each profile slice or portion of each slice ranged from 2.0 to 61.8 cm with

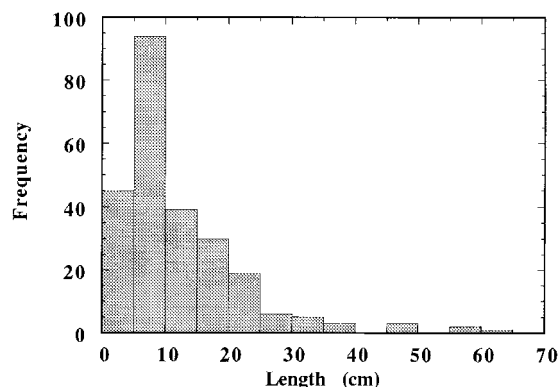


Fig. 6. Frequency distribution of length for macropores.

a mean of 11.6 cm (Fig. 6). Some soil macropores were directly connected with cracks in the bedrock. Water was observed flowing from these cracks during excavation. Most ($\geq 80\%$) visible macropores appeared to terminate in the soil matrix based on spraying colored chalk into the pores.

When the powdered chalk was sprayed into macropores in the O + A horizons, dust emerged from the soil surface. The O + A horizons are thin and porous and have a very high density of roots. This suggests that macropores in the O + A horizon are directly connected to the surface through the matrix of this porous layer. When the powdered chalk was sprayed into macropores in the B horizon, dust sometimes emerged from other macropores in the same soil profile. The specific pathway of this connectivity could not usually be determined by chalk deposition within interconnected macropores. This finding suggests that macropores may also connect through highly porous portions of the soil matrix.

Continuity of soil pipes and macropores is critical to subsurface water movement (e.g., Tsuboyama et al., 1994b). Soil pipes at least 4 to 5 m in length were observed at the experimental forest of the Hokkaido Research Center in Japan (H site) (Kitahara, 1994). Site H is a gentle hillside (13°) with loamy soils. Groundwater rose to high levels and persisted for long durations at site H during the snowmelt season. On the other hand, slope excavations on our much steeper slopes ($>32^\circ$) at Hitachi Ohta revealed that maximum macropore length was 0.6 m. The extent and duration of groundwater rise at the pit in Hitachi Ohta (Noguchi et al., 1992) was much less compared to the Hokkaido

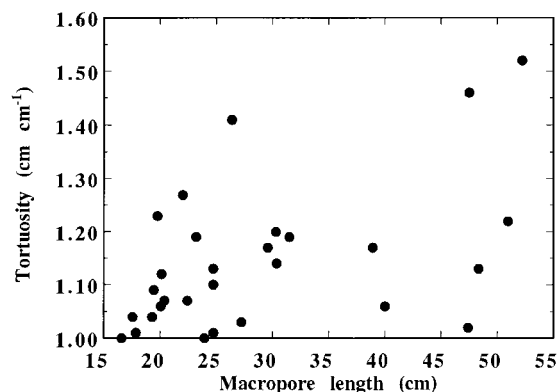


Fig. 7. Relationship between tortuosity and length of macropores.

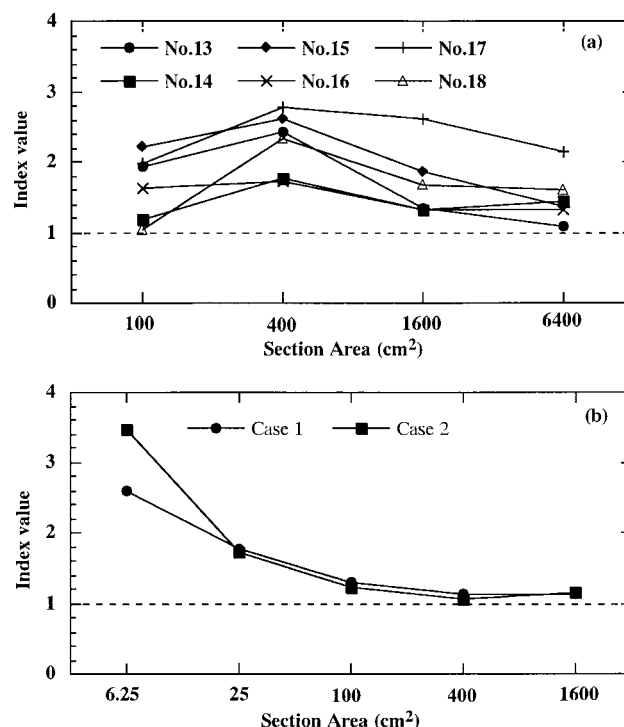


Fig. 8. I-section area relations: (a) macropore distribution; (b) pipe distribution (data provided by H. Kitahara, FFPRI).

results. Subsurface flow is believed to be important in the development and maintenance of soil macropores and pipes (Tsukamoto et al., 1988; Terajima et al., 1996b; Noguchi et al., 1997b). Thus, frequent occurrence of saturated subsurface flow is important in developing longer soil pipes or macropores.

The tortuosity of individual macropores ranged from 1.00 to 1.52 with mean of 1.14. The value of tortuosity tended to increase with increasing macropore length up to ≈ 30 cm. For greater lengths, the tortuosity was quite variable and the sample size was small (Fig. 7). It is reasonable to expect that tortuosity would be low for hydrologically active macropores due to subsurface erosion.

Using a soft x-ray technique, Narioka (1990) measured macropore tortuosities that ranged from 1.2 to 2.0 in a 40 by 70 by 30 mm sample. Kitahara (1989) measured the tortuosity of soil pipes using a plaster impregnation technique and found tortuosities ranging from 1.0 to 1.1 in a sample size of 9.5 by 9.5 by 30 cm. It is impossible to directly compare these results with our tortuosity values because of the different methods and scales. However, these findings suggest the tortuosity may be larger at the small scale (soft x-ray technique) compared to larger scales (plaster impregnation technique and staining technique using powder chalk).

Spatial distribution of macropores using the I_δ index could only be calculated in soil profiles where the number of macropores was 30 or more. Six soil profiles (no. 13–18) met this condition. The I_δ values were highest when they were analyzed for 20 by 20 cm grid cells (400 cm^2) (Fig. 8a). The shape of the I_δ distribution indicated that macropores in the soil profile followed an aggregated distribution with large clumps. Distributions of macropores were significantly different from random

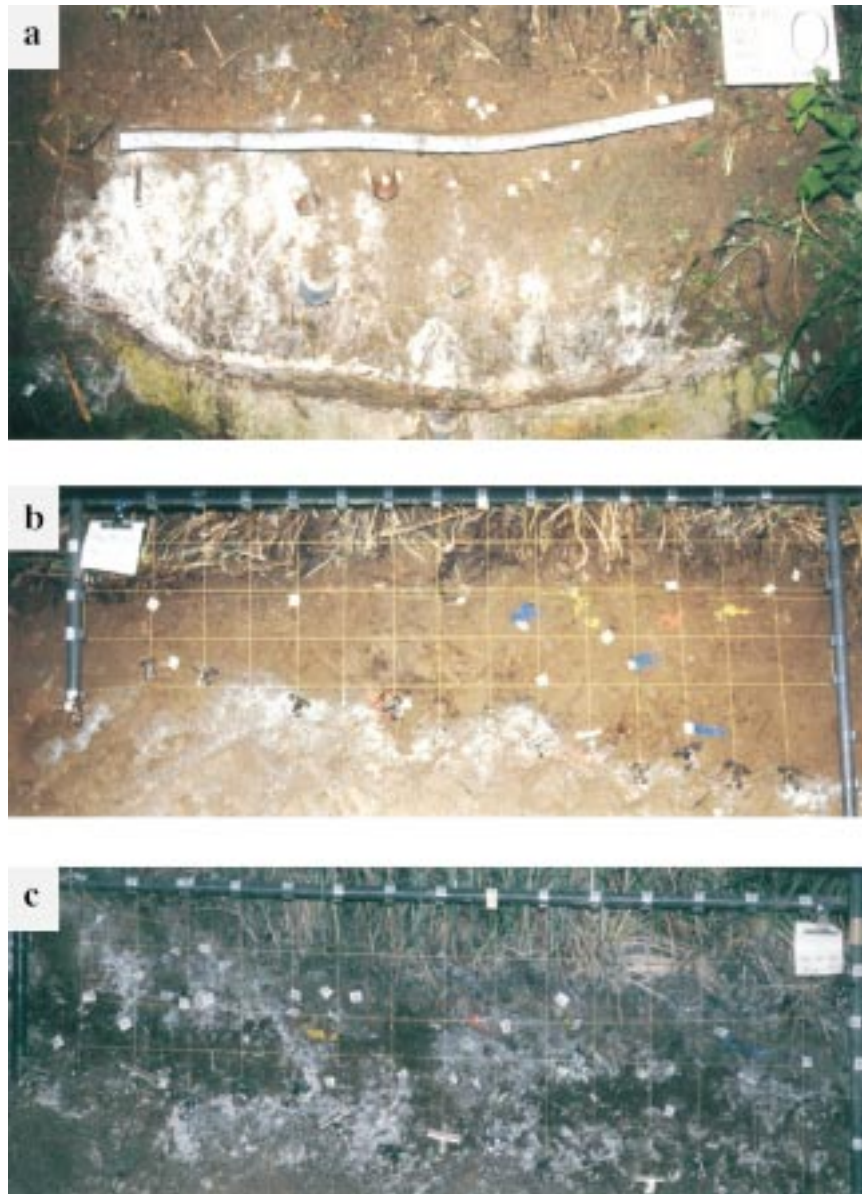


Fig. 9. Distribution of dye (white paint) in the soil profiles: (a) no. 1, (b) no. 11, (c) no. 17.

distributions (i.e., a straight line) based on Eq. [6] for all six soil profiles (no. 15 and 17: $P < 0.01$, no. 13 and 18: $P < 0.025$, no. 14 and 16: $P < 0.1$).

Kitahara et al. (1988) observed that soil pipes in excavated profiles had aggregated distributions with small clumps (Fig. 8b). In Kitahara's study, the soil pipes in the C horizon were hydrologically active. Macropores in our study occurring in the O + A, B and C horizons are both active and inactive. This could explain why the distribution of macropores was different from that of pipes. Nevertheless, both studies indicated that forest soils were composed of heterogeneous media that promoted preferential flow.

Dye Test

The first trace of dye (white paint solution) emerged at the upper left portion in the soil profile 8.7 to 13.8 min after application. Dye did not appear homogeneously on the pit face. Dye distribution concentrated in the left portion of the profile and above certain zones of bed-

rock. Dye also emerged in the surface soil of the upper left portion of the profile (Fig. 9a). The left portion of soil profile was distinctly wetter than the right portion. Dye velocities were calculated as 0.0024 to 0.0038 m s^{-1} . Maximum and average residence velocities from the breakthrough analysis at the same site were 0.0034 and 0.00012 m s^{-1} , respectively (Tsuboyama et al., 1994b). Our measured dye velocities appear to correspond to the high end of the velocity range calculated by Tsuboyama et al. (1994b). They reported that the effective pore volume from the entire profile was much less than the estimate of total volume of pore water, suggesting that preferential flow significantly contributed to subsurface transport of tracer. The dye distribution area was significantly less than the entire soil profile (Fig. 9). Although the dye distribution area does not equal the pore volume, the dye test supports the results from the breakthrough analysis and shows where preferential flow occurred in the soil.

Figure 9b shows the dye distribution at the middle

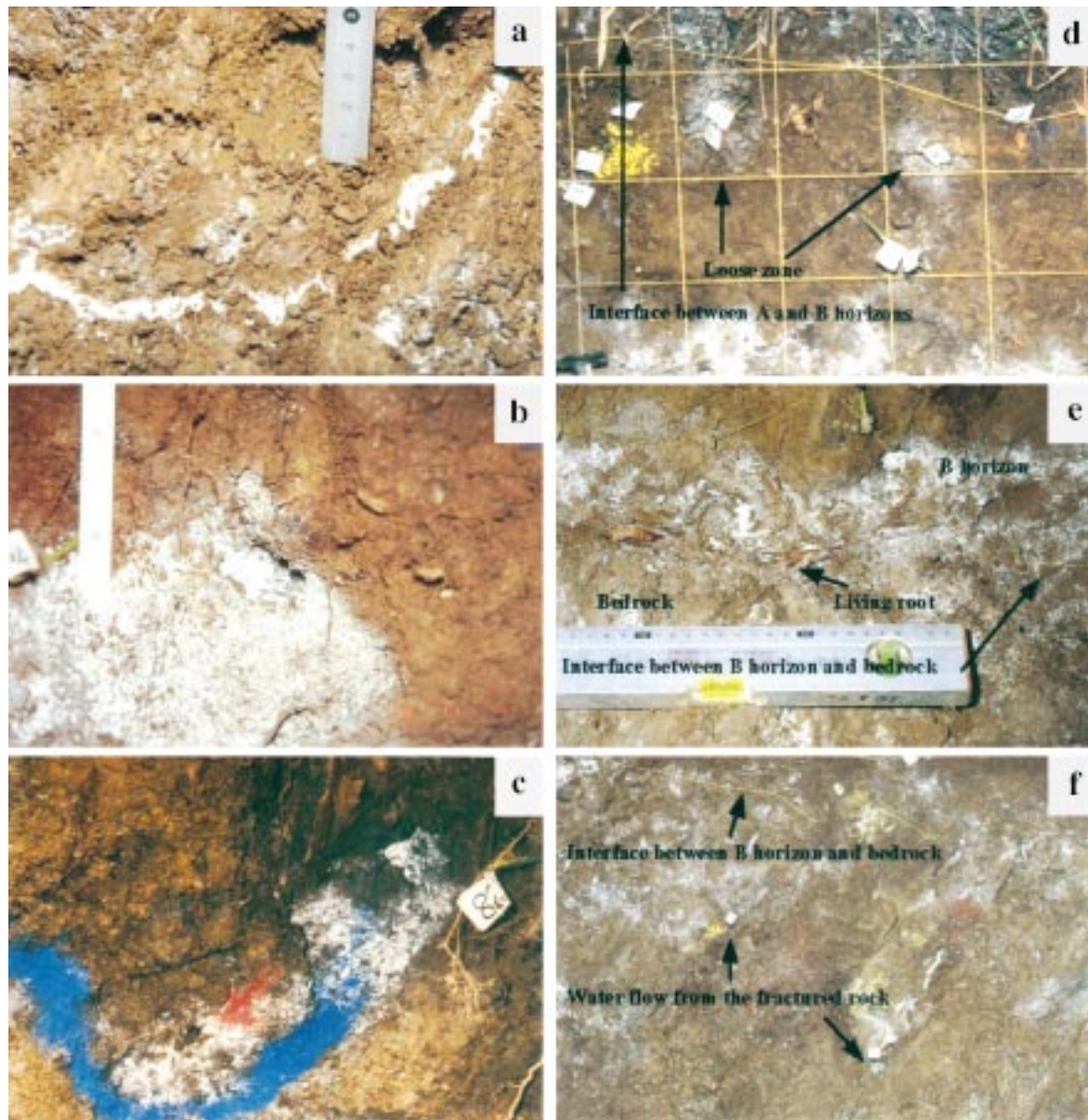


Fig. 10. Various patterns of water movement in the soil as indicated by dye: (a) dye stained along a macropore; (b) stained macropore and soil matrix around it; (c) traced macropore with powder chalk connected stained soil matrix zone; (d) dye distributed vertical loose zone; (e) living roots developed along bedrock; (f) schistosity and water flow from the fractured rock.

cross-section (no. 11) of the soil segment. Dye concentrated in the lower mineral soil horizon just above bedrock. Similar dye distribution patterns were observed in the continuum from soil profile no. 4 through 13. Before we excavated the pit to investigate the dye distribution, discharge including macropore flow from the pit was observed during some storms. Tensiometric heads were also measured. From the observation of tensiometric heads, positive pressure was observed on the bedrock during storms (Noguchi et al., 1992). Though the tensiometric heads were not measured during the dye test, observed discharge (including macropore flow) from the pit face was similar to discharge monitored during natural storm events. These results help confirm that subsurface flow was directed laterally on the bedrock and that a perched water table occurred. Therefore, not

only surface topography but also bedrock topography controls the behavior of subsurface flow. The importance of bedrock topography on subsurface flow has been emphasized in a recent investigations (Wilson et al., 1987; McDonnell et al., 1996; Montgomery et al., 1997). While our study concurs with this general finding, it clearly shows that the small-scale variability in bedrock topography and related flow paths is far too great to be captured by the grid size (2×2 m) used by McDonnell et al. (1996).

Flow patterns in the upper slope portion of the soil profile are shown in Fig. 9c. Near the irrigation line source, the O and A horizons were homogeneously stained but the B horizon was not. This pattern persisted from soil profile no. 20 downslope to profile no. 14 (≈ 50 cm downslope). Subsurface flow was also deflected

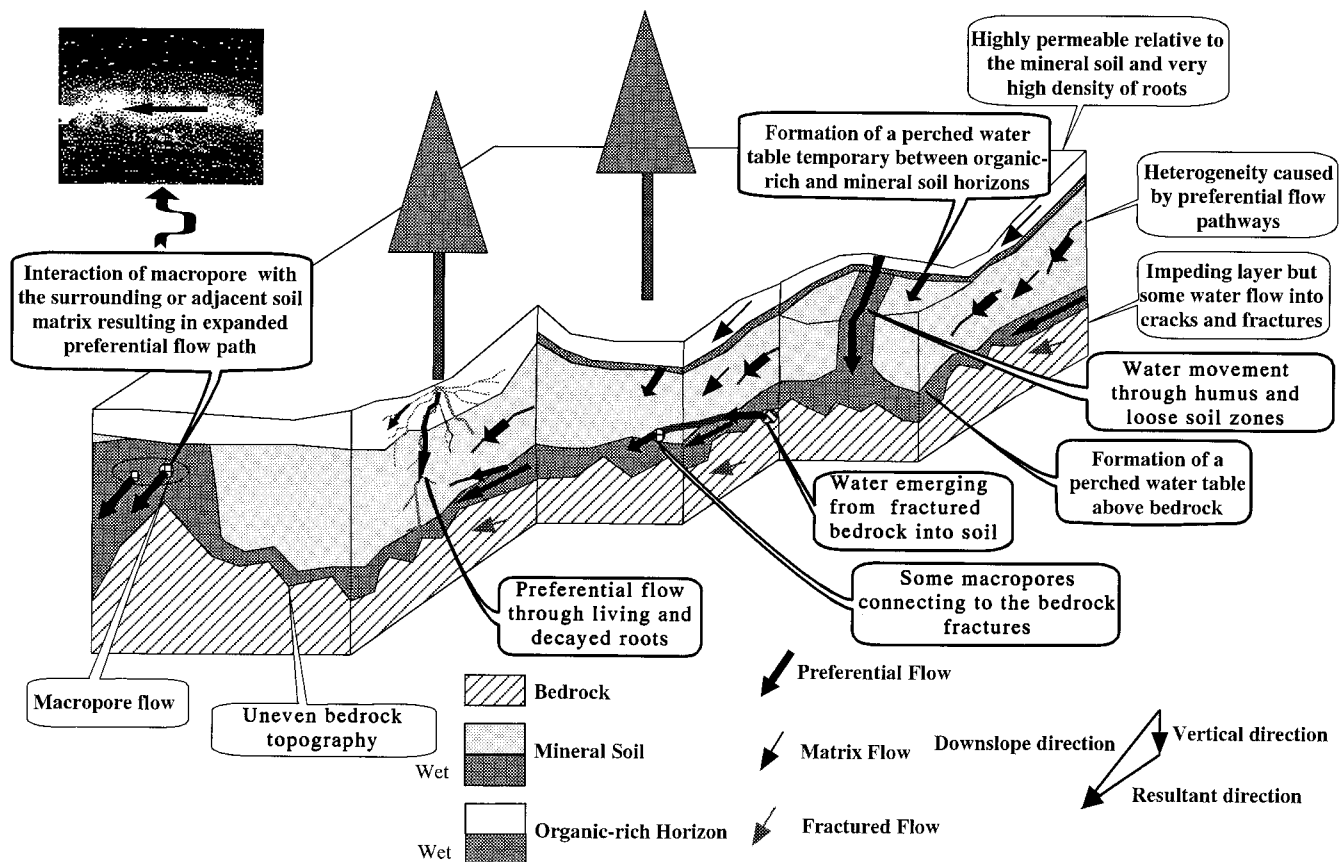


Fig. 11. A schematic-perceptual model of subsurface flow paths in forested slope segment.

laterally between the A and B horizons. Dye distribution patterns were not vertically continuous in the soil profiles. Dye was also distributed in vertical zones of loose soil in the mineral soil horizon (Fig. 10d). The temporary development of a perched water table (as well as associated lateral fluxes) above the B horizon and the hydrologic link caused by buried humus or loose soil are illustrated in the conceptual diagram in Fig. 11.

Various soil water flow patterns were observed in the profiles. Some flow patterns indicated that soil water flowed only through macropores and had little interaction with the soil matrix (Fig. 10a, 11). However, most flow patterns showed a strong interaction between the soil matrix and macropores (Fig. 10b). Interactions between the soil matrix and macropores were suggested in breakthrough results from miscible displacement experiments at the slope segment (Tsuboyama et al., 1994b). The dye distribution patterns observed in this paper lend strong evidence to the flow interactions between macropores and the surrounding soil matrix (Fig. 11).

The soil matrix around the inlets of some discontinuous macropores was stained with white paint (Fig. 10c). This connection suggests that matrix flow was the source of macropore flow. Mizuyama et al. (1994) observed similar results where pipe flow emerged from a gravel layer.

Dye was also distributed vertically in zones of loose mineral soil that were not obvious macropores (Fig.

10d). These zones were a dark brown color similar to the A horizon and had high densities of roots. Omoti and Wild (1979) observed that dye concentrations increased with decreasing dry bulk density of soil. Noguchi et al. (1997a) demonstrated that similar vertical porous zones had hydraulic conductivities 10 to 100 times higher than those in the surrounding mineral soil. Therefore, these more porous zones influenced by living and decayed roots might encourage preferential flow in the vertical direction (Fig. 11). Also, more porous zones could facilitate the slope-parallel connectivity of otherwise discontinuous macropore segments.

Living roots existed at the interface between the B horizon and bedrock (Fig. 10e). Living roots impede flow, but flow can be diverted around the perimeter of roots as evidenced by white paint staining (Noguchi et al., 1997a). If roots penetrate through the entire soil mantle, this pathway around the perimeter of live roots can be an important hydrological linkage to an underlying perched water table (i.e., developed on the bedrock surface) (Fig. 10e, 11). When these roots decay, the pathway may expand and become a larger and more conductive macropore. On this site, about 70% of the macropores in the organic-rich layer and 55% in the B horizon were estimated to be formed by root channels (Noguchi et al., 1997b). White paint stains were observed in and around both living and decayed roots as well as loose vertical zones caused by decayed roots. Thus, it appears that both living and decayed root sys-

tems contribute to preferential flow pathways in this forest soil.

The surficial geology at this site is metamorphic rock, primarily schist and amphibolite. The fractures in the schist strike 46° E and dip 43° SE. Considerable water flow from the fractured rock was observed during excavation (Fig. 10f). Dye distribution was observed along the fractures within the bedrock (Fig. 10f). A portion of this flow in the fractured bedrock would represent infiltrating soil water. The return flow from the bedrock caused the localized wetness in the left portion of the soil profile (Fig. 9a). While the flow on the fractured rock does not necessarily follow surface topography, the flow within the bedrock fractures appears somewhat independent of both surface and bedrock topography, especially for the small scale at which flow paths were observed at our site. Thus, it is important to investigate bedrock fracture pattern as well as bedrock topography.

SUMMARY AND CONCLUSIONS

Morphological characteristics of macropores and related preferential flow pathways in a forested slope segment were mapped using powdered chalk tracers and a liquid white paint. Most described macropores were short (2–62 cm) and straight with cross-sections approximating ellipses. Macropores tended to be aggregated in large clumps in the soil profile. About 70% of the macropores were discontinuous within an 8-cm soil section. Although individual macropores were ≤ 62 cm, they were connected into longer preferential flow paths by: (i) interaction with the surrounding or adjacent soil matrix; (ii) contact with living and decayed roots; (iii) formation of a perched water table above bedrock, and between A and B horizons; (iv) interaction with water moving through bedrock fractures; (v) interconnections with humus in the organic horizon; and (vi) influence of surface and bedrock topography (Fig. 11).

Both living and decayed roots and associated vertical zones of friable soil contribute to preferential flow pathways in this forested slope. In some cases, soil water flowed only through macropores without interacting with the soil matrix. The more highly connected patterns of preferential flow involved an interaction between the soil matrix and macropores. Such field evidence lends support to the idea of expansion of macropores by interaction with surrounding mesopores (Luxmoore and Ferrand, 1993; Tsuboyama et al., 1994b).

Subsurface flow was deflected laterally over the bedrock and a perched water table developed. Dye distribution in fractures of the bedrock indicate that zones of localized wetness were generated by fracture flow diverting some of the infiltrated water. Based on these results, we present a further development of our concepts related to flow within a hillslope segment at Hitachi Ohta (Fig. 11).

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